

# JEE MAIN REVISION MASTER: CHEMICAL BONDING AND MOLECULAR STRUCTURE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 26 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                     | Frequency | Category           |
|---------------|--------------------------------------------------|-----------|--------------------|
| Very High     | VSEPR Theory, Hybridization & Molecular Geometry | 17 PYQs   | Molecular Geometry |
| High          | Molecular Orbital Theory & Dipole Moments        | 9 PYQs    | Bonding Theories   |

## 2. Master Formula Reference Card

| Concept / Topic                                  | Mathematical Formulation & Governing Equations                                                                                  |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| VSEPR Theory, Hybridization & Molecular Geometry | $\text{Steric Number} = \text{Bond Pairs} + \text{Lone Pairs}$ , $H = \frac{1}{2}(V + M - C + A)$                               |
| Molecular Orbital Theory & Dipole Moments        | $\text{Bond Order} = \frac{N_b - N_a}{2}$ , $\text{Bond Order} \propto \text{Bond Energy} \propto \frac{1}{\text{Bond Length}}$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 VSEPR Theory, Hybridization & Molecular Geometry (Very High Yield — 17 PYQs)

**Core Principle:** VSEPR theory: Repulsion order: lp-lp > lp-bp > bp-bp. Hybridization steric number  $H = \frac{1}{2}(V + M - C + A)$ . SF<sub>4</sub>: sp<sup>3</sup>d, 1 lp, see-saw. ClF<sub>3</sub>: sp<sup>3</sup>d, 2 lp, T-shaped. XeF<sub>2</sub>: sp<sup>3</sup>d, 3 lp, linear. XeF<sub>4</sub>: sp<sup>3</sup>d<sup>2</sup>, 2 lp, square planar. PCI<sub>5</sub>: sp<sup>3</sup>d, trigonal bipyramidal (axial bonds are longer and weaker than equatorial bonds).

■ **Caution / JEE Pitfall:** In PCI<sub>5</sub>, the 2 axial P-Cl bonds are longer and weaker than the 3 equatorial bonds due to greater repulsion from equatorial electron pairs.

■ **Speed Trick:** In sp<sup>3</sup>d hybridization, lone pairs ALWAYS occupy EQUATORIAL positions to minimize 90° repulsions (e.g. ClF<sub>3</sub> has both lone pairs in equatorial positions implies T-shape).

**Representative Source PYQs:**

- [2026 Shift] Q#3 (MCQ): "The types of hybrid orbitals of nitrogen in NO<sub>2</sub><sup>+</sup>, NO<sub>3</sub><sup>-</sup> and NH<sub>4</sub><sup>+</sup> respectively are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The change in hybridization (if any) of the 'Al' atom in the following reaction is AlCl<sub>3</sub> + Cl<sup>-</sup> → AlCl<sub>4</sub><sup>-</sup>..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "Bond enthalpies of A<sub>2</sub>, B<sub>2</sub> and AB are in the ratio 2:1:2. If bond enthalpy of formation of AB is -100 kJ mol<sup>-1</sup>. The bond en..." [Ans: N/A]

### 3.2 Molecular Orbital Theory & Dipole Moments (High Yield — 9 PYQs)

**Core Principle:** MOT: Bond Order =  $(N_b - N_a) / 2$ . For  $\leq 14$  electrons (extB\_2, extC\_2, extN\_2):  $\pi 2p_x = \pi 2p_y$  fills before  $\sigma 2p_z$ . For  $> 14$  electrons (extO\_2, extF\_2):  $\sigma 2p_z$  fills before  $\pi 2p$ . O<sub>2</sub> has 2 unpaired electrons in degenerate antibonding  $\pi^* 2p$  orbitals implies paramagnetic with Bond Order = 2. Dipole moment  $\mu = q \times d$ ; symmetrical molecules (extCO\_2, BF\_3, CCl\_4) have  $\mu = 0$ .

■ **Caution / JEE Pitfall:** Liquid O<sub>2</sub> is attracted into a magnetic field because it is PARAMAGNETIC (a fact that Valence Bond Theory completely failed to explain, but MOT successfully explains).

■ **Speed Trick:** If Bond Order = 0, the diatomic molecule does not exist (e.g. He<sub>2</sub>, Be<sub>2</sub>).

**Representative Source PYQs:**

- **[2026 Shift] Q#1 (MCQ):** "With respect to resonance structures of CO<sub>3</sub><sup>2-</sup> ion, which of the following statements are correct? (a) All C-O..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "In which of the following option/options, the order of arrangement does not agree with the variation of property indicated against it? (a) \$..." **[Ans: N/A]**
- **[2025 Shift] Q#1 (MCQ):** "The electronic configuration of X and Y are given below:  $\begin{aligned} &X: 1s^2 2s^2 2p^6 3s^2 3p^3 \\ &Y: 1s^2 2s^2 2p^6 3s^2 \end{aligned}$  & Y:  $1s^2 2s^2 2p^6 3s^2$ ..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

With respect to resonance structures of CO<sub>3</sub><sup>2-</sup> ion, which of the following statements are correct? (a) All C-O bonds in CO<sub>3</sub><sup>2-</sup> are equivalent (b) There are three resonance structures possible for CO<sub>3</sub><sup>2-</sup> ion (c) The position of carbon and oxygen should change in every resonance structure (d) The formal charge on carbon atom is -2

- (A) a, b and c
- (B) a and b only
- (C) b and d only
- (D) a, b and d

**Correct Answer: None**

**Solution Analysis:** In the carbonate ion, CO<sub>3</sub><sup>2-</sup>, carbon is at the centre and the three oxygen atoms are attached to it. The ion has three possible resonance structures. In each structure, one C=O double bond and two C-O<sup>-</sup> single bonds are shown. The double bond can be placed with any one of the three oxygen atoms. So statement (b) is correct. In resonance, the actual structure is not any one of these structures. The actual carbonate ion is a resonance hybrid. Because of this, all three C-O bonds become equivalent. Each C-O bond has

**Benchmark Problem 2: KCET 2026 (Q#3)**

The types of hybrid orbitals of nitrogen in  $\text{NO}_2^+$ ,  $\text{NO}_3^-$  and  $\text{NH}_4^+$  respectively are

- (A)  $\text{sp}^3$ ,  $\text{sp}$  and  $\text{sp}^2$
- (B)  $\text{sp}$ ,  $\text{sp}^2$  and  $\text{sp}^3$
- (C)  $\text{sp}^2$ ,  $\text{sp}$  and  $\text{sp}^3$
- (D)  $\text{sp}^2$ ,  $\text{sp}^3$  and  $\text{sp}$

**Correct Answer: None**

**Solution Analysis:** For finding the hybridisation of nitrogen, we count the number of electron domains around nitrogen. Electron domains include: sigma bonds lone pairs For  $\text{NO}_2^+$  : Nitrogen is bonded to two oxygen atoms. Structure is:  $\text{O}=\text{N}=\text{O}$  Nitrogen has 2 sigma bonds and no lone pair. So, steric number:  $2+0=2$  Steric number 2 means linear shape and  $\text{sp}$  hybridisation. So, nitrogen in  $\text{NO}_2^+$  is  $\text{sp}$  hybridised. For  $\text{NO}_3^-$  : Nitrogen is bonded to three oxygen atoms. Due to resonance, all three N-O bonds are equivalent. Nitrogen has 3 sigma b